Step 1 : Define key elements

Planning horizon : 3 stages, representing the 3 countries

Decision epochs T : $T=1,2,3$ corresponding to Country 1, 2, 3 respectively

State variable S_t : the number of medical teams remaining to be allocated when entering the stage t

Decision variable x_t : the number of medical teams allocated to Country t

Transition function $f_t(S_t, x_t)$: $S_{t+1} = f(S_t, x_t) = S_t - x_t$

Contribution function $C_t(S_t, x_t)$: the additional person-years of life for Country t for the allocation of medical team x_t

Objective function Z : the maximum of additional person-years of life in total

$$Z = \sum_{t=1}^{3} C_t(S_t, x_t)$$

Step 2 : Recursive relation

Let $V^*(S_t)$ be the maximum effectiveness from stage t to 3

$$V_t^*(S_t) = \max\{C_t(S_t, x_t) + V_{t+1}^*(S_{t+1})\}$$
$$= \max\{C_t(S_t, x_t) + V_{t+1}^*(S_t - x_t)\}$$

Set boundary condition : $V_4^*(S_4) = 0$

Step 3 : Backward induction

① Stage 3 : $V_3^*(S_3) = \max\{C_3(S_3, x_3) + V_4^*(S_4)\}$
$$= \max\{C_3(S_3, x_3)\}$$

$x_3 = S_3$

$V_3^*(0) = C_3(0,0) = 0$, $V_3^*(1) = C_3(1,1) = 50$, $V_3^*(2) = C_3(2,2) = 70$

$V_3^*(3) = C_3(3,3) = 80$, $V_3^*(4) = C_3(4,4) = 100$, $V_3^*(5) = C_3(5,5) = 130$

② Stage 2: $V_2^*(S_2) = \max\{C_2(S_2, x_2) + V_3^*(S_3)\}$

$$= \max\{C_2(S_2, x_2) + V_3^*(S_2 - x_2)\}$$

$V_2^*(0) = C_2(0,0) + V_3^*(0) = 0$, $x_2^* = 0$

$$\left.\begin{aligned} &C_2(1,0) + V_3^*(1) = 50 \\ &C_2(1,1) + V_3^*(0) = 20 \end{aligned}\right\} \Rightarrow V_2^*(1) = 50,\ x_2^* = 0$$

$$\left.\begin{aligned} &C_2(2,0) + V_3^*(2) = 70 \\ &C_2(2,1) + V_3^*(1) = 70 \\ &C_2(2,2) + V_3^*(0) = 45 \end{aligned}\right\} \Rightarrow V_2^*(2) = 70,\ x_2^* = 0 \text{ or } 1$$

$$\left.\begin{aligned} &C_2(3,0) + V_3^*(3) = 80 \\ &C_2(3,1) + V_3^*(2) = 90 \\ &C_2(3,2) + V_3^*(1) = 95 \\ &C_2(3,3) + V_3^*(0) = 75 \end{aligned}\right\} \Rightarrow V_2^*(3) = 95,\ x_2^* = 2$$

$$\left.\begin{aligned} &C_2(4,0) + V_3^*(4) = 100 \\ &C_2(4,1) + V_3^*(3) = 100 \\ &C_2(4,2) + V_3^*(2) = 115 \\ &C_2(4,3) + V_3^*(1) = 125 \\ &C_2(4,4) + V_3^*(0) = 110 \end{aligned}\right\} \Rightarrow V_2^*(4) = 125,\ x_2^* = 3$$

$$\left.\begin{aligned}
C_2(5,0)+V_3^*(5) &= 130\\
C_2(5,1)+V_3^*(4) &= 120\\
C_2(5,2)+V_3^*(3) &= 125\\
C_2(5,3)+V_3^*(2) &= 145\\
C_2(5,4)+V_3^*(1) &= 160\\
C_2(5,5)+V_3^*(0) &= 150
\end{aligned}\right\} \Rightarrow V_2^*(5)=160,\ x_2^*=4$$

② Stage 1: $V_1^*(S_1) = \max\{C_1(S_1, x_1) + V_2^*(S_2)\}$

$$= \max\{C_1(S_1, x_1) + V_2^*(S_1 - x_1)\}$$

$$\left.\begin{aligned}
C_1(5,0)+V_2^*(5) &= 150\\
C_1(5,1)+V_2^*(4) &= 170\\
C_1(5,2)+V_2^*(3) &= 165\\
C_1(5,3)+V_2^*(2) &= 160\\
C_1(5,4)+V_2^*(1) &= 155\\
C_1(5,5)+V_2^*(0) &= 120
\end{aligned}\right\} \Rightarrow V_1^*(5)=170,\ x_1^*=1$$

the Optimal solution: $x_1^*=1,\ x_2^*=3,\ x_3^*=1$

$$Z = V_1^*(5) = 45+125+0 = 170$$